A
PROJECT REPORT
ON

2 Player Chess

By

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B.Tech CE Semester-VI
SUBJECT: SYSTEM DESIGN PRACTICE

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CERTIFICATE

This is to certify that the project entitled "**2 Player Chess"** is a Bonafide report of the work carried out by

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Date:

ABSTRACT

The main purpose of this **2 Player Chess** game is to play a chess game between two players. The main functionality of this project is that two players can play chess with each other and chat at the same time. Another major functionality is that the player can also play against the bot. While the player can also see the dot of possible moves. Any player can resign at any time. Both players will have their own backward clock. When any player's time expires, the opponent will be the winner. When any player checkmate his opponent, he is considered the winner. Second, a win can be earned when one player resigns the game. There is also a feature of pawn promotion in the game. Pawn promotion occurs when a pawn reaches the farthest rank from its original square—the eighth rank for White and first rank for Black. When this happens, the player can replace the pawn for a queen, a rook, a bishop, or a knight.

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1. Introduction

1.1 Brief Introduction

The Aim of this online web-based chess project is to make two players play against each other. Generally, the chess game is normally played against the human user and the computer on which the user is operating, so the need is to make the chess game being played against two human users on two different machines on the network, no matter their physical location.

1.2 Technology/platform/Tools Used

- Socket.io
- NodeJS

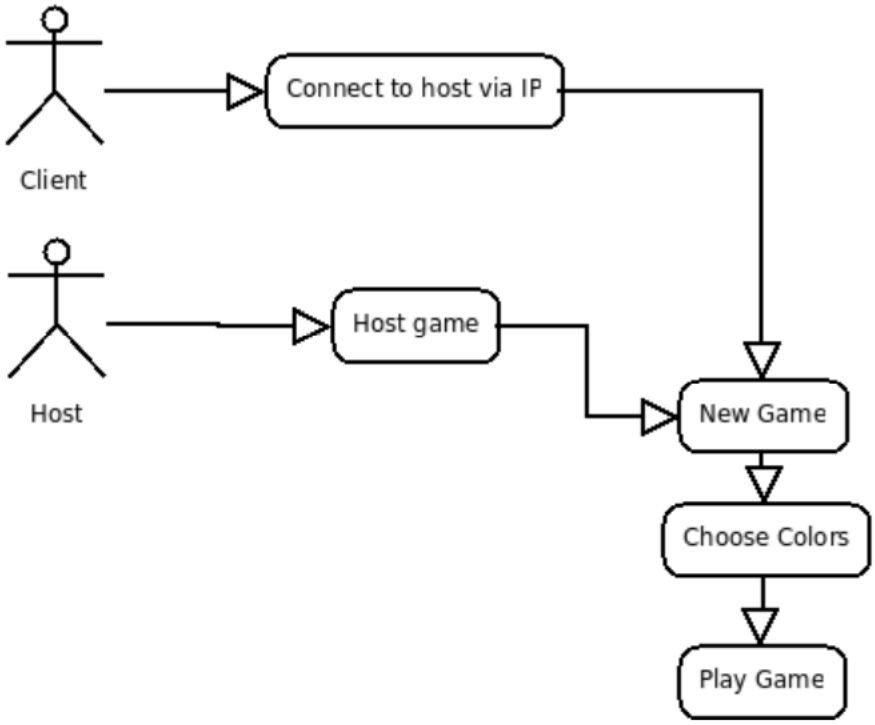
1.3 Product Scope

User can also play against bot. The user can play against his friend. The user view valid moves. Both users can chat with each other. Any user can resign at any time.

2. Connect

Main Flow: User opens browser and specifies the link of the user to whom he/she wants to connect (i). If the connection attempt is successful the players start a new game (ii). The players choose who will play black and who will play white (iii). The board is drawn so that each player has his/her pieces at the bottom of his/her display (iv). The game moves to the play game state with the white player as the active player (v).

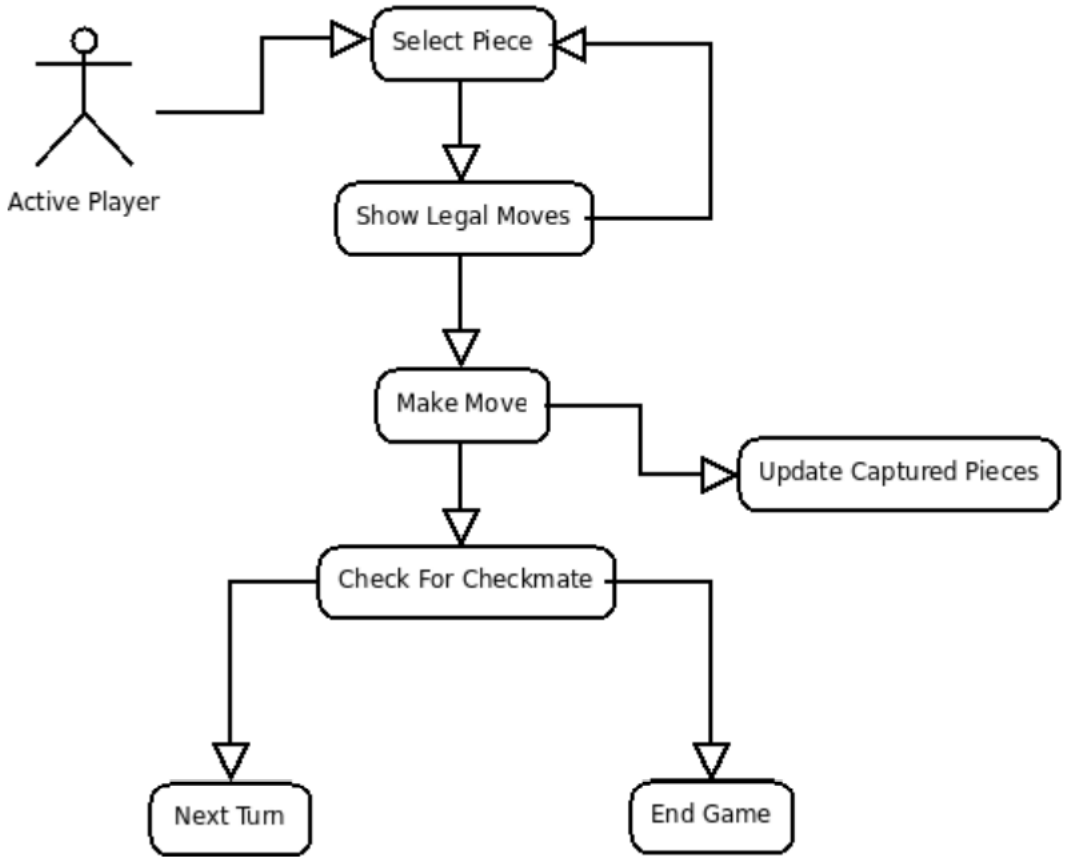
SubFlows: I. User opens browser and connects to an opponents via link. II. Once there is a successful connection a new game is started. III. The players argue over color. IV. Each player sees the game board from the appropriate perspective. V. The game moves into the play game state with the white player as the active player.



Precondition: During the Play Game state

Main Flow:
The active player clicks a piece to select it. The game displays the positions it can move to. The player selects the new destination by clicking. The piece is moved there if it is a valid move. Their opponent becomes the active player.

Alternate Flow:
The active player may decide to select a different piece by clicking one of their own. If there are no valid moves and the active player is not in check the game ends as a stalemate. If there are no valid moves and the active player is in check the inactive player wins.



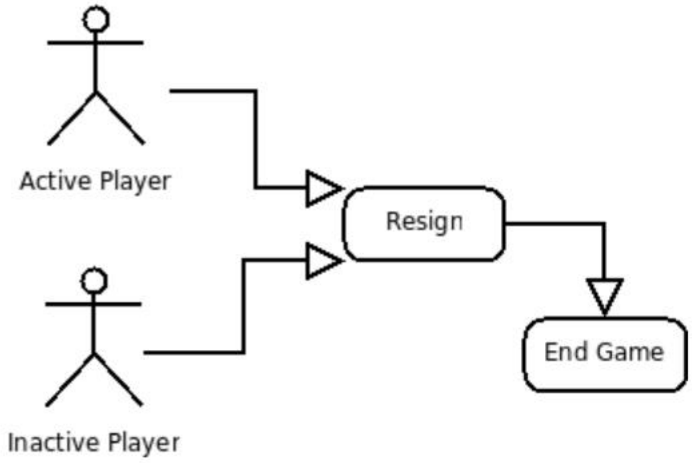
Preconditions: During the Play Game state

Main Flow:
At any time during gameplay, a player selects the forfeit option (i). The player then confirms their forfeit (ii). The game is immediately ended with the forfeiting player as loser (iii).

Sub Flows:

I. A player chooses to forfeit
II. The same player confirms their forfeit
III. That player loses and the game is over.

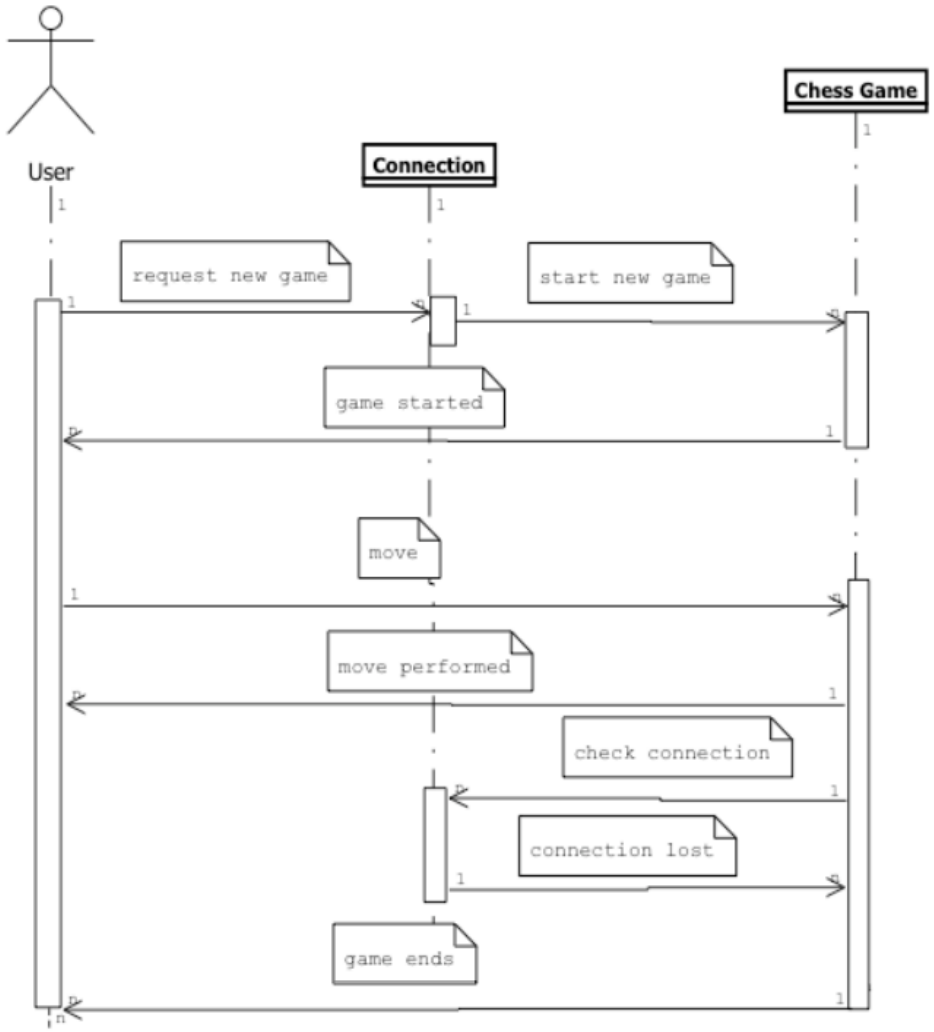
Alternative Flow: The player decides not to confirm the forfeit. Game player resumes as usual



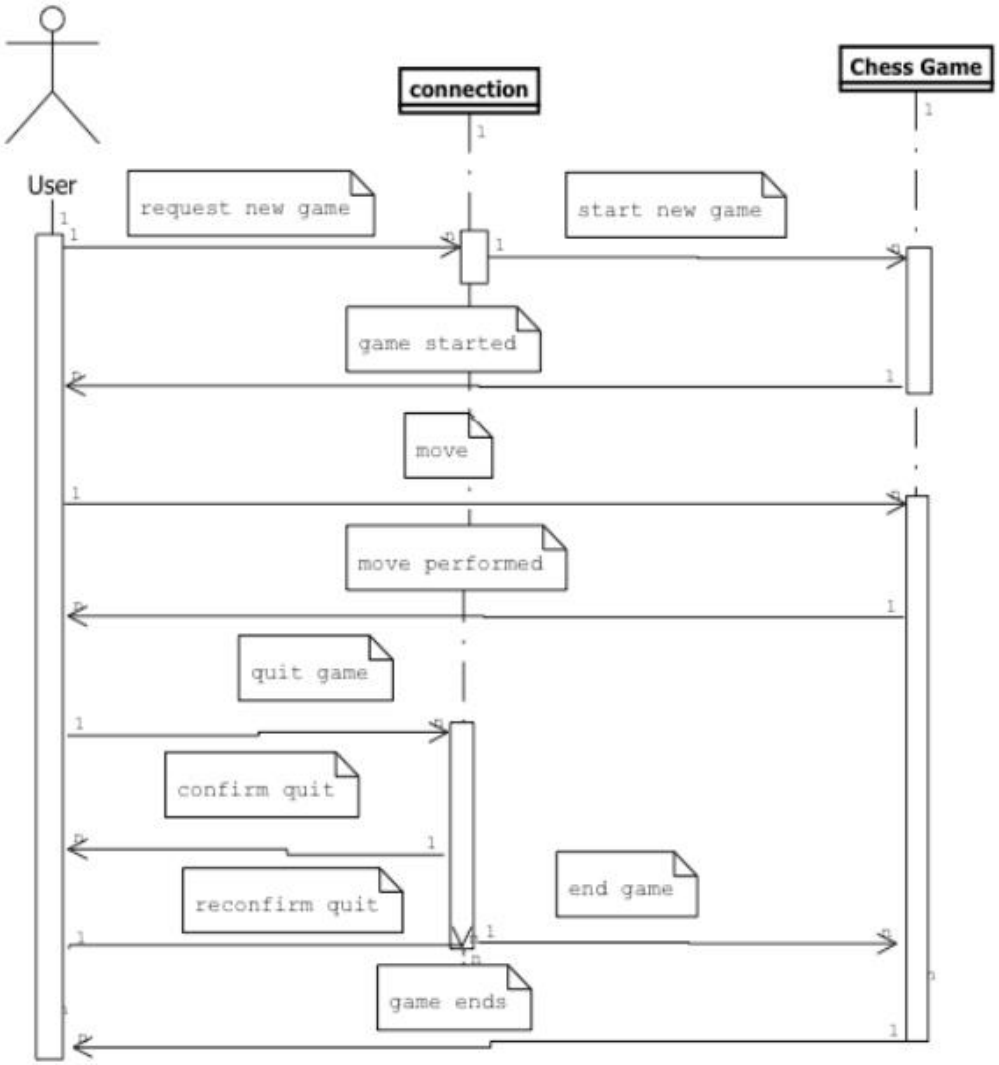
3. Sequence Diagram

3.1 Sequence Diagram

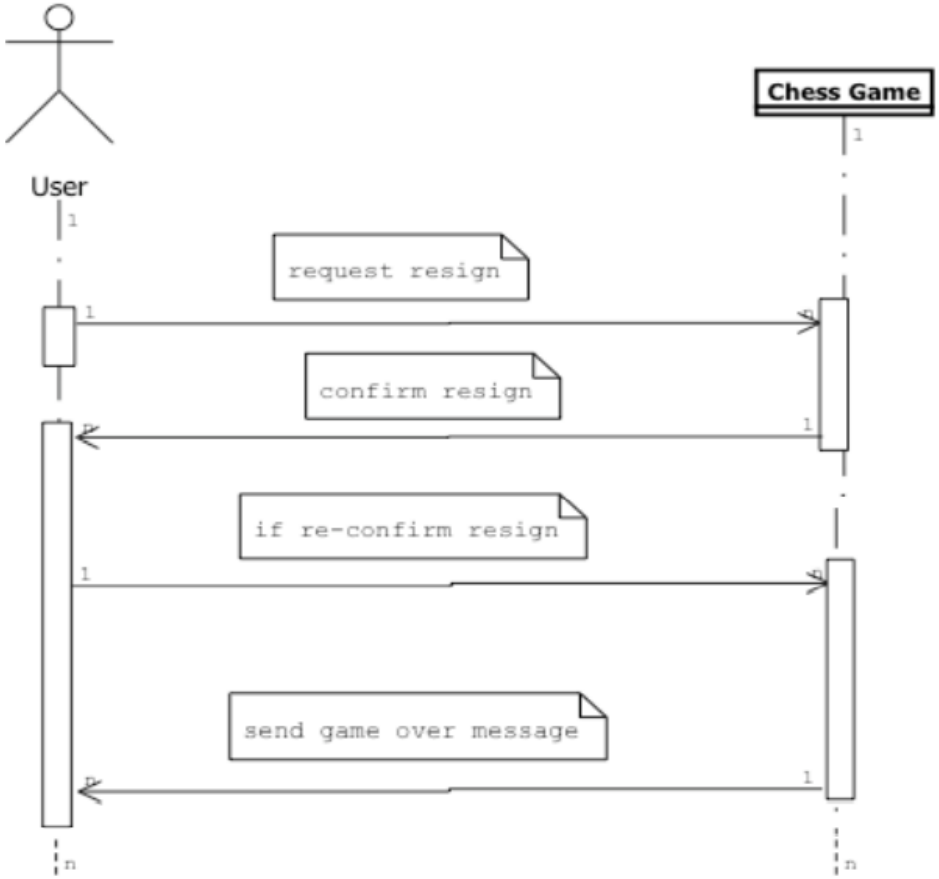
3.1.1. Play Game Sequence Diagram



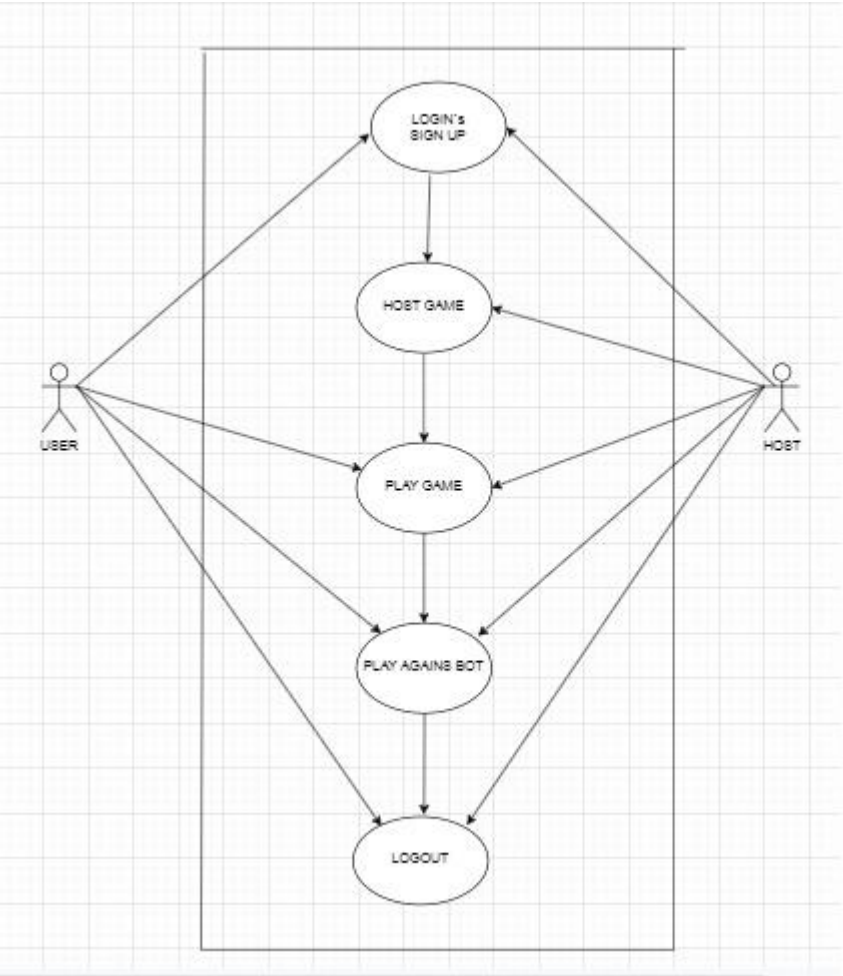
3.1.2. Sequence diagram for play game and quit in the middle



3.1.3. Sequence diagram for Resign and confirm

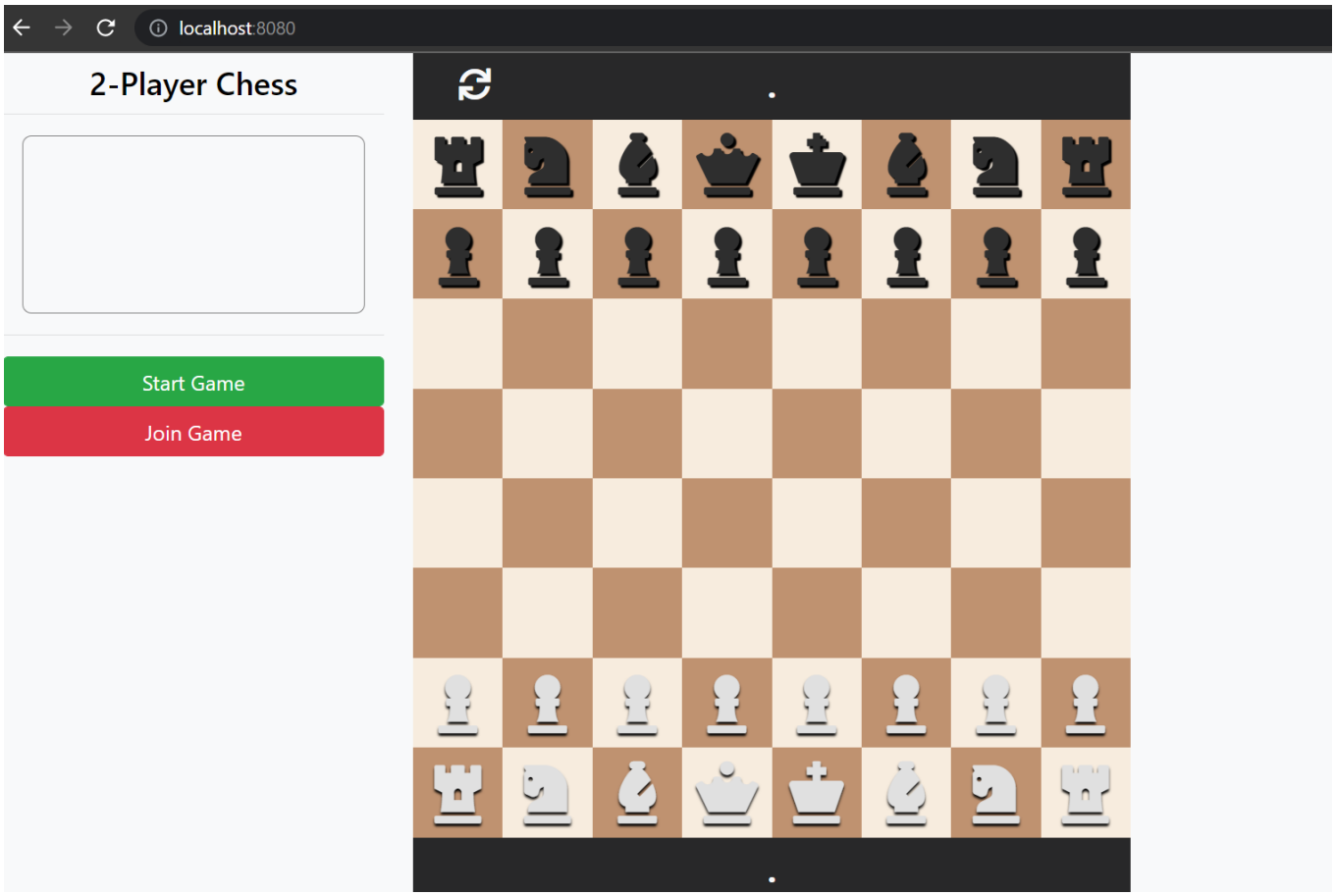


4. Use-Case Diagram

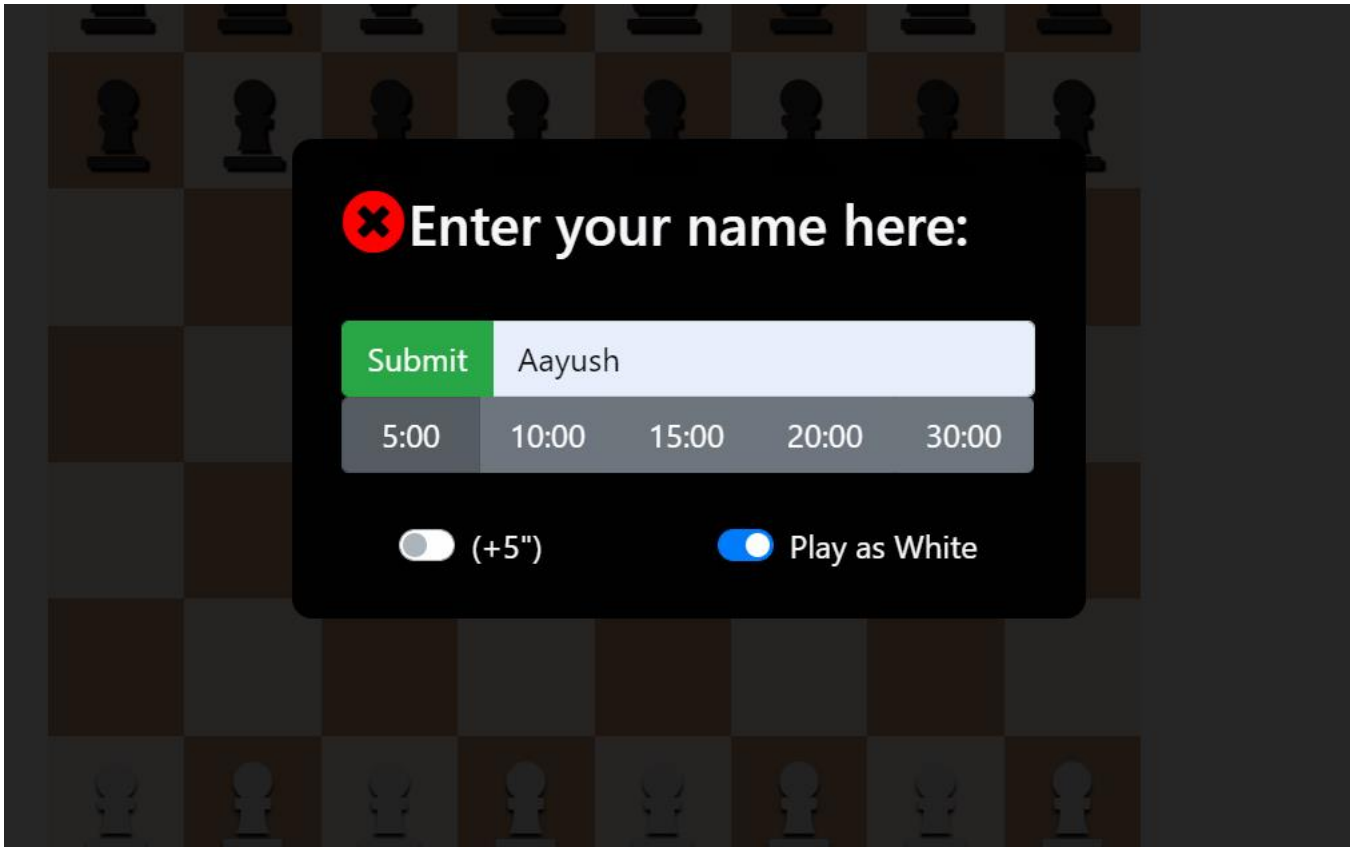


5. Screenshot

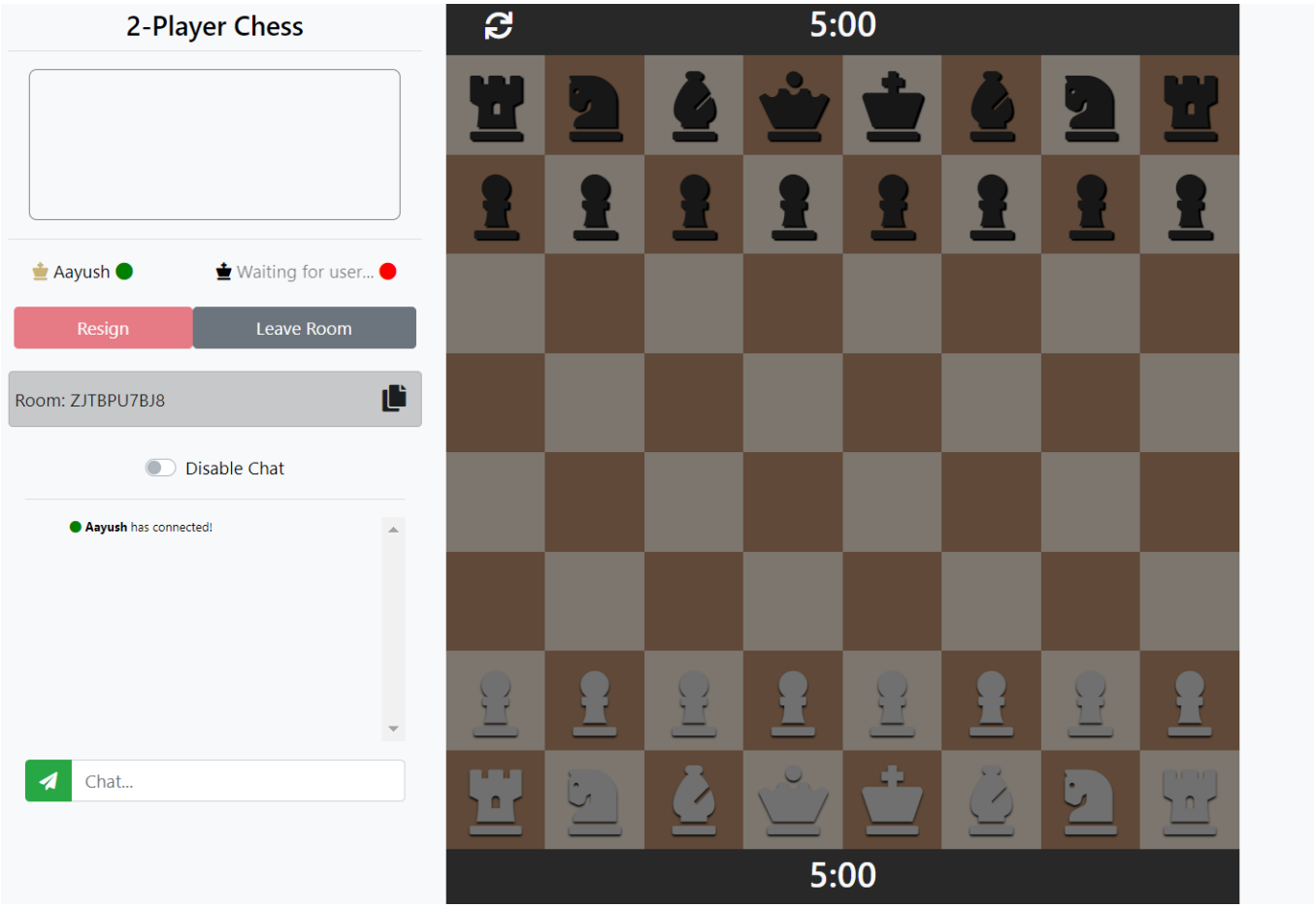
1. **Home Page**



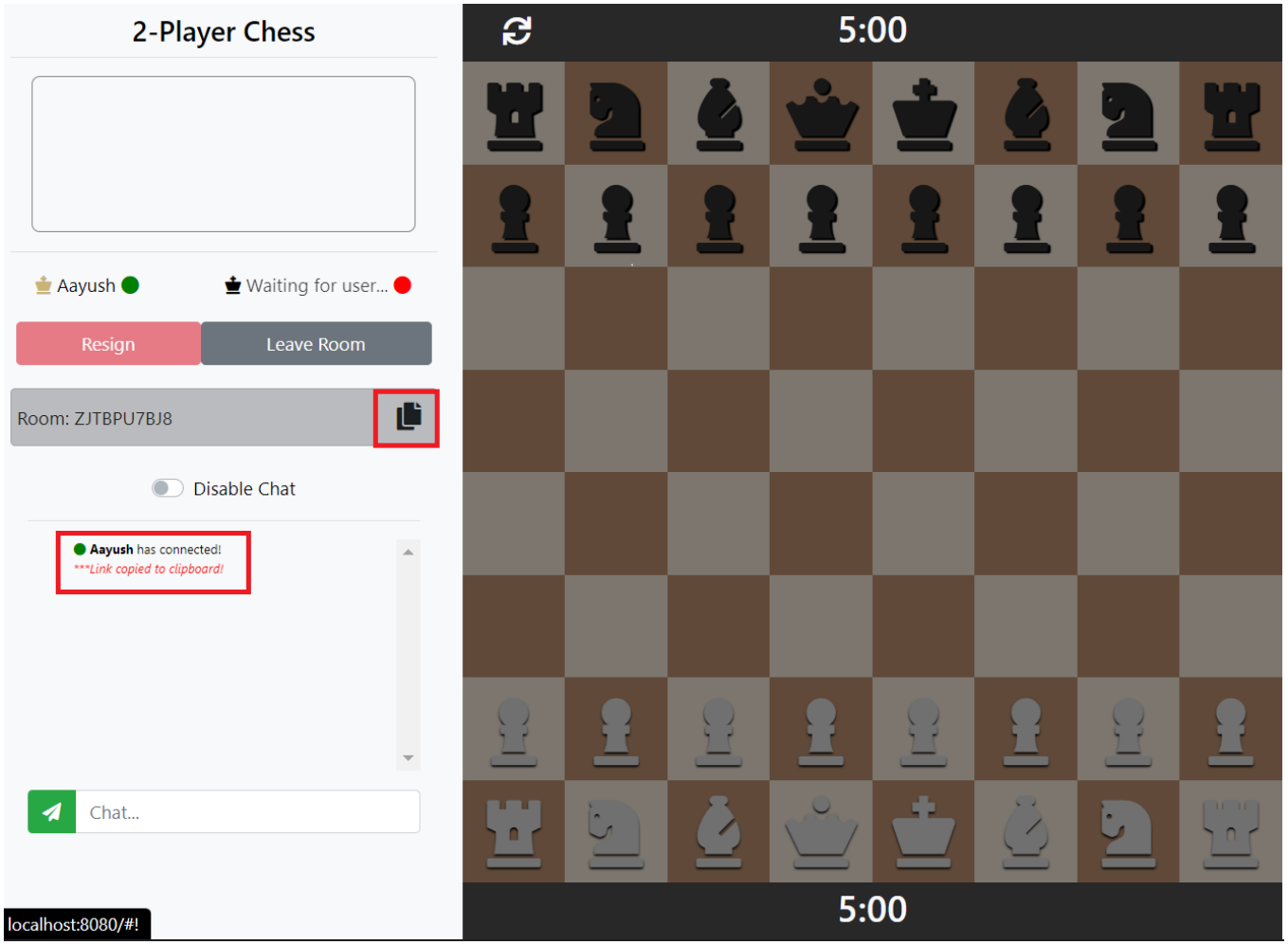
2. Start Game


Enter your name here:
Submit
Aayush
5:00
10:00
15:00
20:00
30:00
(+5")
Play as White

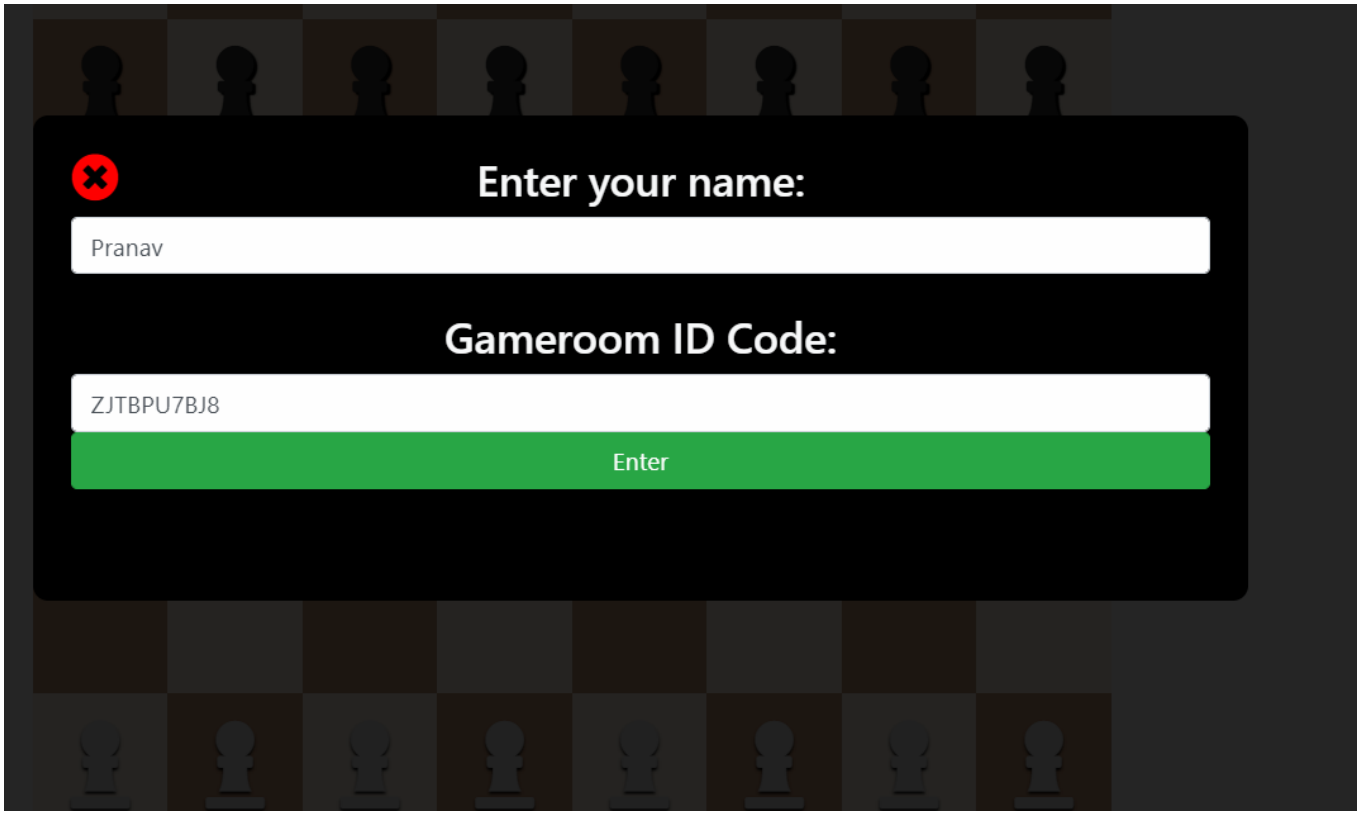
3. Game started at user 1 side:



4. Link copied:

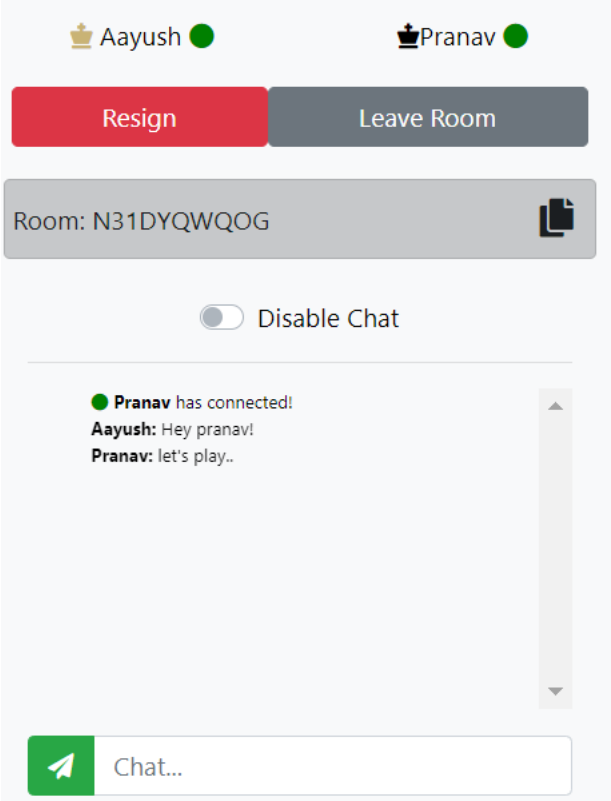


5. Opponent Enter Game Room ID Code:

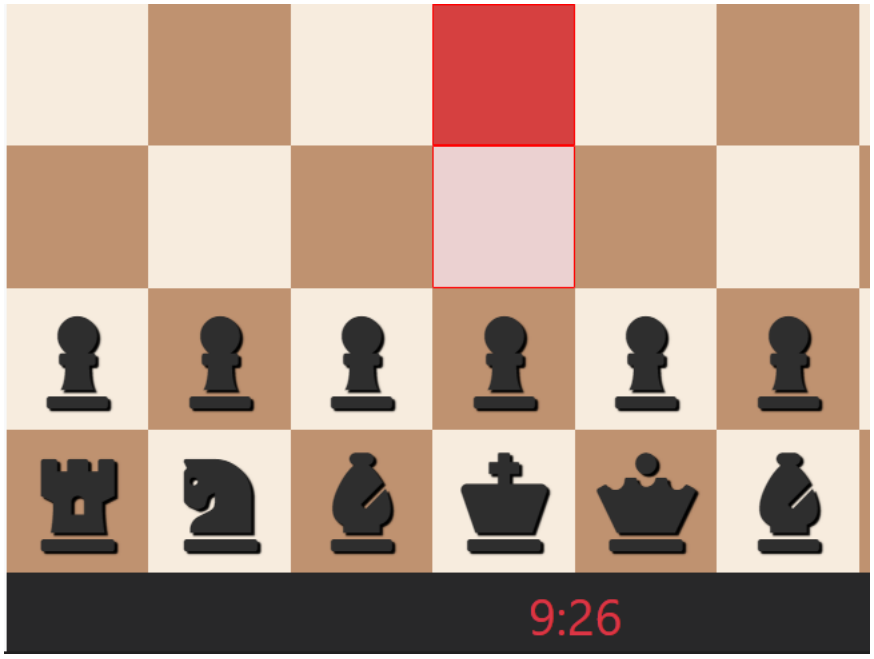


6. Opponent Joined and Game Started:

7. They are communicating:



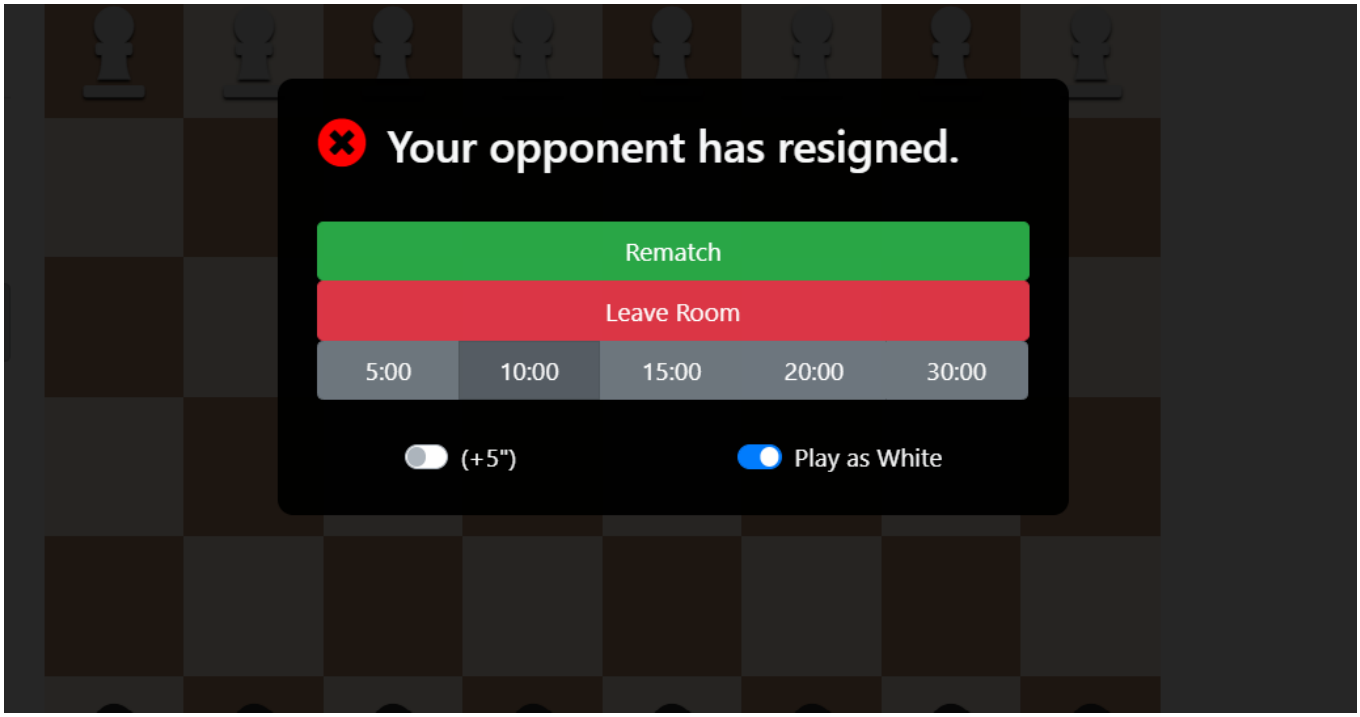
8. Possible Moves Dots:



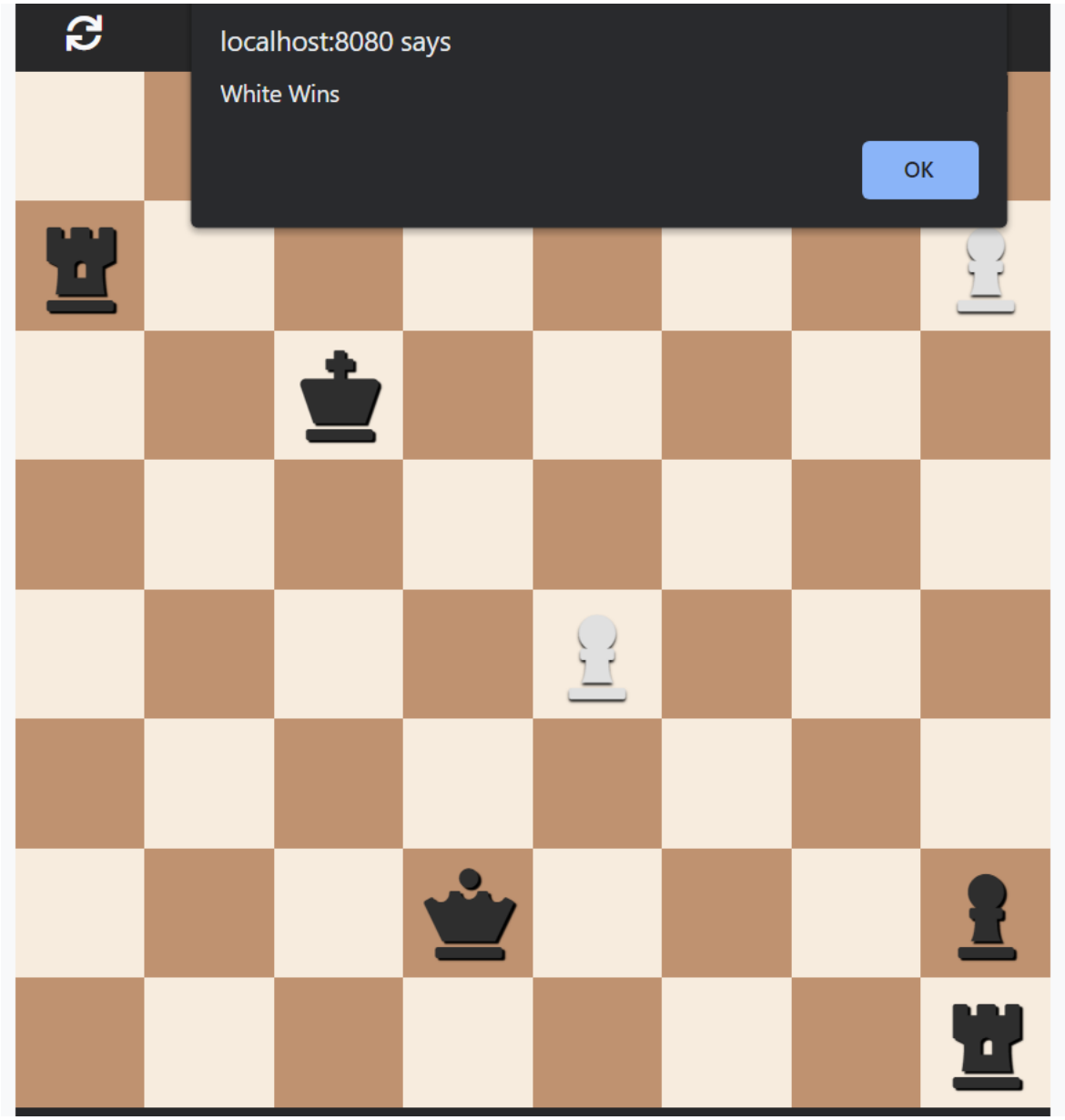
9. Past Moves Will Be Visible:

10. Promoting pawns when reaches at other side:

11. Anyone User Resigned:

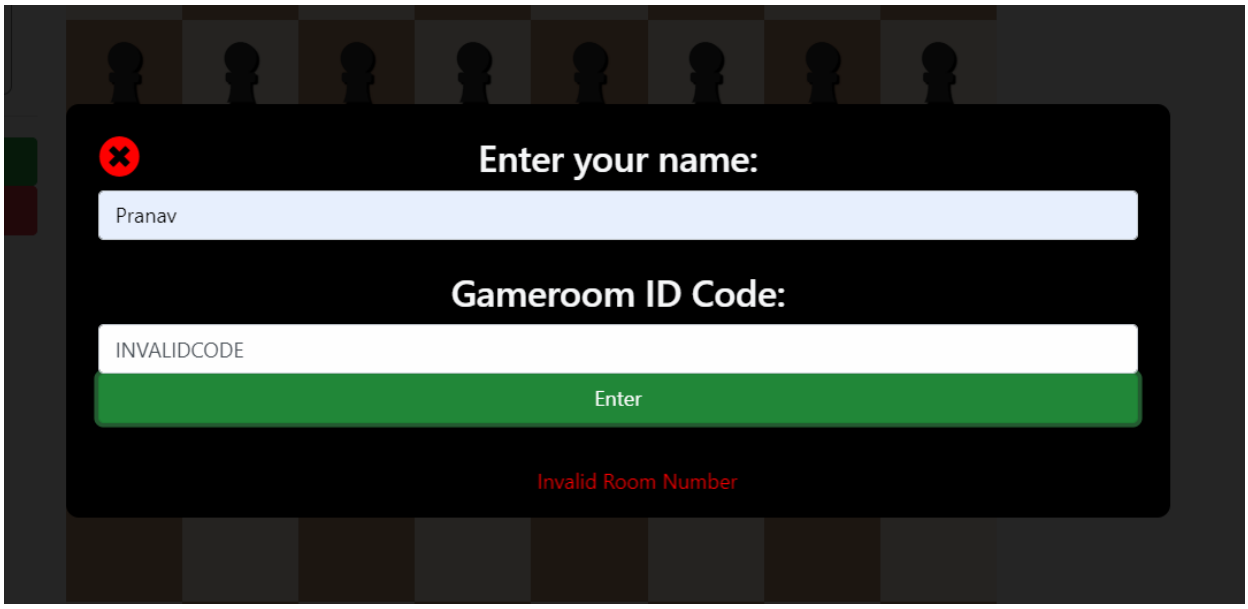


12. Black's time is up and White wins:

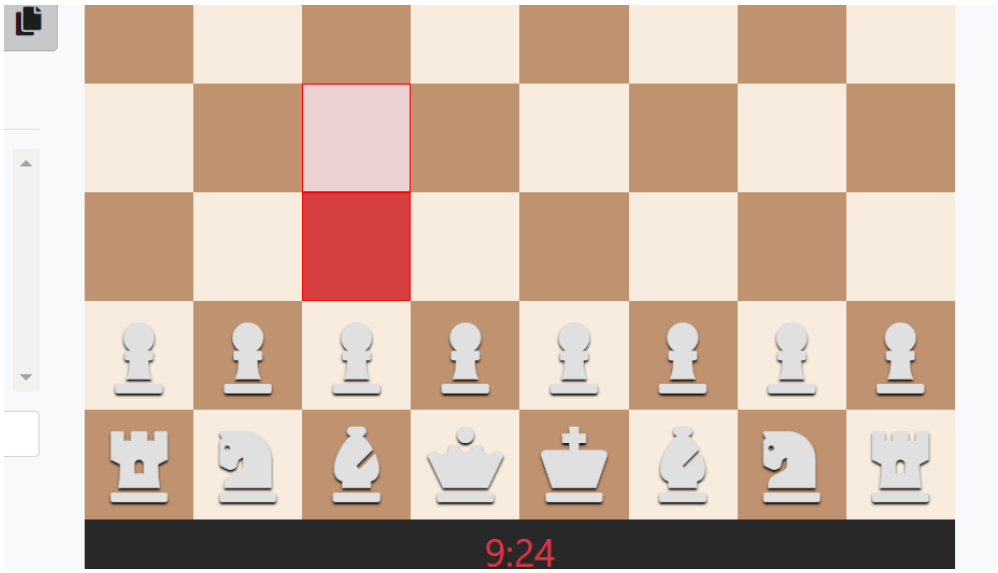


6. Test Cases

1) When an opponent enters incorrect code, it is considered invalid and the user cannot enter the room.



2) The user can only make valid moves.



3) When one player stops a game, the other player's screen is grayed out and that player is notified.

7. Conclusion

The functionalities are implemented in system after understanding all the system modules according to the requirements. Functionalities that are successfully implemented in the system are:

- Two players can play with each other.
- They can chat with each other.
- Any user can Resign or Leave the room at any time.
- Both users have their own separate timer.
- The user can see the legal move dots on the board.

After the implementation and coding of system, comprehensive testing was performed on the system to determine the errors and possible flaws in the system.

8. Limitation and Future Extension

Limitation:

- The user cannot create his own account so he cannot store his own record
- Users can't play against bots for now.
- Sometimes there can be a connection problem in between.

Future Extension:

- Once the website is hosted the user will be able to play against the bot.
- The user will also be able to register and Login.
- There will also be a chess tournament.

9. Bibliography

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- https://nodejs.org/en/docs/
- https://socket.io/get-started/chat
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